Digital developments for reliable ZE-store deliveries

Research and implementation proposal. 19 October 2026. This report supports the three-slide SA3 presentation and the Lesson 4 requirements (slides 39-44). The pilot has not been run. The register entries and interfaces below are proposals; no council decisions or signatures are claimed (*Lesson 4: Digital developments*, 2026, slides 39-44).

Team: Made Bhisma Sayta Nugraha; Quinten Poppinga; Francesco Gatteschi; Sander van de Pol; Scholte van Ringelestijn.

Decision and hypothesis

Outbound proposes a limited pilot of delivery-window-aware planning in its existing TMS, supported by live ETA monitoring. The decision is whether to extend these functions after a feasible pilot improves delivery reliability without harming other deliveries or cost per cage. Earlier planned timestamps must translate into feasible changes to departure times or stop order.

Hypothesis: scheduling selected ZE-store deliveries earlier inside their existing agreed windows, with realistic service times and operational constraints, may reduce late arrivals. The data identifies a planning-margin issue; it does not establish the cause of each delay or guarantee an OTIF improvement.

Submission without C3/C10: submit this conditional proposal now. Desk analysis can proceed; store selection and live changes await verified access and subcontractor restrictions (*Board archive index*, 2026).

Observed evidence from the supplied data

Measure	ZE stores	Other stores
Deliveries	3,847	15,395
Punctuality, supplied on_time flag	87.47%	98.38%
Window length	30 min	60 min
Planned-arrival margin before closing	15 min	45 min
Mean actual minus planned arrival	+7.14 min	+10.46 min

Note. Authors' calculations from *Store replenishment* (n.d.), supplied in General (1).zip, Outbound/store_replenishment.csv: 19,242 distinct delivery IDs, 120 stores, 14 October 2024 to 11 January 2025. Margin = window_end minus planned_arrival. ZE stores have less scheduling margin even though their mean deviation from plan is smaller. ZE deliveries represent approximately 66% of flags marked not on time.

Interpretation: test scheduling margins first. The data does not establish vehicle scarcity or the cause of individual delays; no improvement is guaranteed.

The two digital developments

Development 1: delivery-window-aware semi-automated planning

Extend or configure the existing Routiq TMS to consider store windows, cage volume, store/load-specific service estimates, vehicle suitability and access constraints. Compare feasible earlier arrival positions within the existing windows. A planner reviews the proposed route and departure plan before release. Use historical medians by store/load group first; advanced AI is not required. Configurable service times and windows are documented capabilities, not evidence that Routiq already licenses them (PTV Logistics, n.d.-a, n.d.-b).

Implementation: audit licensed TMS functions and validate historical data now. Compare earlier scheduling in desk/shadow plans within existing windows; keep unverified access constraints marked unknown. Obtain C10 and approvals before selecting routes for a live pilot; configure only verified constraints and release only executable changes. Orderpicking receives the approved loading/departure sequence through the existing WMS-TMS connection. No picking-performance intervention is proposed.

Evidence and maturity: Holland et al. (2017), UPS ORION, reports real production use; PTV technical documentation demonstrates configurable service times and time windows. Neither establishes results for FreshWay. UPS serves parcels and differs from multi-temperature store replenishment. The paper is peer reviewed, but that alone does not establish Tier 1 accountable financial-report evidence. PTV documentation is a capability source, not independent outcome evidence.

Development 2: live ETA monitoring with planner alerts

Use existing board-computer position/progress data, exported via the paid vendor API, and compare ETA with the delivery window. Alert Outbound planners to an approaching miss. The planner can contact the store or approve a safe, feasible route adjustment; an alert alone does not prevent a late delivery. Record action and outcome for evaluation.

Implementation: confirm API licence, fields, frequency and service conditions; map vehicle/route/delivery IDs; agree a pilot update frequency with the supplier; connect the TMS plan and existing BI environment; test missing/stale-data alerts; train planners. A nightly BI load cannot support live alerts. Use an operational feed/console rather than assuming the current overnight warehouse is real time.

Evidence and maturity: Descartes Systems Group (n.d.) reports Stewart Foodservice’s execution visibility. A dated trade article reports Cube Route adoption in early 2004 (“The Value of Information,” 2005). This corroborates historical production, not a current Descartes version. Neither source establishes a FreshWay benefit. Today’s API, compatibility and costs require supplier confirmation; no historic gains or prices are transferred.

Costs and who benefits

Outbound owns the pilot and receives operational benefits; IT supports integration and API access. Potential additional costs cover configuration, integration, API access, training and support. Amounts are to be confirmed through supplier quotations. Count only additional charges beyond existing contracts; no price or savings is assumed. The TMS contract expires on 31 March 2027. The case software payback requirement is 3-5 years (*Board card C11: IT roadmap and contracts*, 2026; *Company dossier*, 2026, sec. A4).

Architecture and daily use

Retain one company ERP, one WMS and one TMS. The WMS-TMS API already exists (*Company dossier*, 2026, sec. A5). Planning is an enhancement within the TMS; monitoring is a function using its plan and existing board computers. No additional departmental TMS is proposed. The presentation shows the simplified DFD; the detailed payloads below define the proposed flows, retaining the documented Orderpicking and Inventory handovers (*Department file: Orderpicking*, 2026; *Department file: Outbound transport*, 2026).

Sender / receiver	Information and event	Owner / status
Orderpicking to TMS planning	Delivery/store/site IDs; window; required cages by temperature.	Orderpicking owns normal load inputs. Confirm the existing field mapping.
TMS to Orderpicking	Approved route/loading sequence and planned departure; update after a released plan revision.	Outbound owns plan. Extend existing API only as needed.
TMS to driver	Route ID, delivery IDs, stop order, vehicle assignment and planned arrivals; at dispatch or approved revision.	Outbound planner approves release.
Board computer/vendor cloud to monitoring	Vehicle/route ID, timestamp, position and execution status; paid API at agreed frequency.	Outbound/IT confirm supplier rights, fields and latency.
Monitoring/planner to TMS	Delay-risk alert followed by an approved adjustment; rejected options logged too.	Outbound planner owns action. No autonomous diversion.
Execution confirmation to Inventory & SCM	Delivery/store IDs, confirmed delivered cages/cases and actual arrival; on completion/correction.	Outbound owns record. Confirm quantity/POD source; GPS is insufficient.
Historical records to planning/review	Store/load unload times and planned/actual arrival deviations.	D1 export supports analysis, not a claimed live database.

Interface Agreements for new cross-department flows

Draft IA-OP: sender Orderpicking, receiver Outbound; agree required quantities, delivery/store/window keys and release/correction events. Draft IA-SCM: sender Outbound, receiver Inventory & SCM; agree actual delivery confirmation and required-quantity reconciliation. Proposed initial format: UTF-8 CSV with ISO timestamps; production API mapping, time zone, ownership, correction handling and latency must be confirmed. Each receiver signs only after agreement. No signatures or implemented fields are invented.

Obtain charging constraints from Storage (*Board card C05: Energy and grid capacity*, 2026). Vehicle sharing is outside scope.

Pilot design and KPI evaluation

Work now; conditional four-week operational pilot

Available now: validate historical data, compare earlier schedules and request supplier details. Do not treat unverified access or contract conditions as satisfied.

Conditional week 1: validate timestamps/flags, operational constraints, available fields and API terms. Obtain C10 for candidate stores; C3 if subcontractor timings or arrangements change. Record actual departure times prospectively. Verify vehicle suitability, driver hours and Storage charging constraints.

Week 2: run shadow plans and establish a comparable baseline. Select a small feasible route group; do not change contractual delivery windows. Predefine the comparison, practical success threshold, acceptable cost and service limits with the Board. Choose sample size/duration from baseline variability and the improvement worth detecting; extend the trial if evidence is insufficient.

Weeks 3-4: operate approved plans on comparable or randomly allocated route-days where feasible. Match site, weekday, window and cage demand; consider shared fleet effects. Log departure/arrival, early arrivals, unloading, alert timestamps, planner actions, delivered quantities, distance and all costs. Monitor other stores served by changed routes.

Company baselines and targets below are stated case figures, not results recalculated from D1 (*Department file: Outbound transport*, 2026). The ZE punctuality baseline is calculated from *Store replenishment* (n.d.).

Measure	Baseline / company target	Evaluation
ZE punctuality	87.47% supplied flag; no ZE-specific norm provided	Primary pilot outcome. Reconcile flag rule first; compare matched groups and report counts, percentage-point change and uncertainty.
Early arrivals and late minutes	Measure baseline from agreed timestamp rule	Guardrail: earlier planning must not shift failures into early arrivals or unsafe waiting.
OTIF	Case baseline 97.2%; target 98.5%	Only calculate when required-versus-delivered quantities are available. Do not label arrival punctuality OTIF.
Cost per delivered cage	Case ~EUR 7.73; target EUR 7.00	Include distance/time, API, integration and recurring costs on comparable scope.
Cages per route	Case 45.4; target 52	Service/capacity guardrail; verify actual vehicle capacities rather than assuming every truck holds 66.
Backhaul route share	Case 12%; target 30%	Not directly targeted; no improvement claim.

Scale/stop decision: expand only after reliable data shows a meaningful punctuality gain within agreed cost, safety, early-arrival and other-store limits. An underpowered pilot is inconclusive. No gain is promised.

Register entries and collaboration

Two proposed Technology Register rows

Entry	Source / date / maturity	FreshWay application
OB-SA3-01: window-aware planning	Holland et al., 18 Jan 2017, DOI below: documented UPS production. PTV docs, n.d., accessed 19 Oct 2026: commercial capability. Academic production report, not verified Tier 1 financial evidence.	Existing TMS enhancement; first shadow test, then limited pilot. KPIs: punctuality/OTIF hypothesis; cost and cage utilisation guardrails.
OB-SA3-02: live ETA monitoring	Descartes Systems Group (n.d.) and “The Value of Information” (2005). Trade coverage reports Cube Route adoption in early 2004. Historical production / Tier 2 deployment evidence; no current-version claim or audited forecast.	Existing board-computer feed plus paid API and planner alerts. Measure alert lead time and actions; service benefit unproven.

These entries are drafted here for the Technology Officer to post in the company register. Posting, Architecture Board approval and receipt confirmations have not been performed by this document.

Other departments’ developments: information unavailable

Documented dependency: Storage owns charging constraints (*Board card C05: Energy and grid capacity*, 2026). C05 establishes a dependency, but does not identify a selected digital development. The assignment requests a relevant partner citation (*Lesson 4: Digital developments*, 2026, slide 44).

At the time of submission, no relevant digital development selected by another department was available to cite. Cross-department integration will be assessed when those selections are shared.

Before hand-in and before pilot launch

Completion actions: post both register entries, confirm architecture and interfaces, and add a relevant partner-development citation when that information becomes available. The manager records the commitment and submits the Board Memo. These actions are not claimed complete.

Before launch: review C10, and C3 if subcontracted routes change; obtain supplier quote and Board approval. These missing cards limit implementation, not submission of this proposal.

Data limitations that constrain the recommendation

D1 lacks departure timestamps and required quantities (*Store replenishment data dictionary*, n.d.). Resolve 58 boundary-flag mismatches and inconsistent consecutive-stop timings before testing. Vehicle capacities and eligibility remain unverified. C3/C10 may change feasibility; the historical period does not establish current operating conditions.

AI transparency

ChatGPT supported data analysis, source research, reasoning, drafting and visual production. This was substantial assistance. The team must validate calculations and claims, complete collaboration steps and use the course-approved declaration; no completed team review is claimed.

References

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Reference note. No author is explicitly credited on the supplied internal case documents, so titles occupy the author position. Dates use supplied document metadata; no dataset release date is invented. C3 and C10 are not in the reference list because their contents were not available. The archive index supports only their titles and the decision dependencies. APA 7 author-date citations and hanging-indent references are used within the FreshWay report design.